PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Software Test Plan

Smart Complaint Management System
Version 1.0
Prepared: 06-10-2026
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1. Introduction

1.1 Purpose

Defines testing strategy, test items, environment, security validation, test cases, traceability, risks and reporting for the Smart Complaint Management System.

1.2 Scope

Authentication, complaint submission/validation, complaint viewing/tracking, admin access control, management, status/resolution, persistence, security, performance and usability.

1.3 References

- SCMS SRS v1.0
- SCMS SAD v1.0
- SE Mini-Project Deliverables Part-1, PES University, 2026
- Supplied PES University Test Plan template

2. Test Items

- Registration/authentication
- Complaint submission
- User dashboard/tracking
- Administrator dashboard
- Status/resolution management
- Authorization/security
- Database/API integration

3. Features to be Tested

Requirement	Feature
FR-01/02	Registration and authentication
FR-03/04	Complaint submission and validation
FR-05/06	User viewing/tracking
FR-07	Admin access control
FR-08/09/10	Admin management/status/resolution
FR-11	Persistence
NFR-01	Performance
NFR-03/04	Security/authorization
NFR-06	Usability

4. Features Not to be Tested

- Third-party infrastructure internals outside team control.
- Production-scale traffic beyond evaluation environment.
- Features not included in the agreed SRS.

5. Test Approach / Strategy

- Unit testing
- Integration testing
- System/end-to-end testing
- Acceptance testing
- Functional positive/negative testing
- Security testing

- Performance testing
- Usability testing

5.1 Security Validation

- Passwords hashed and not returned.
- Protected endpoints reject unauthenticated requests.
- Normal users cannot access admin endpoints.
- Malformed inputs are safely validated.
- No secrets committed/exposed.

5.2 Entry Criteria

- Build starts successfully.
- Database/schema available.
- Test users include normal user and admin.
- Relevant APIs implemented.

5.3 Exit Criteria

- All planned cases executed.
- No critical defect open.
- High-priority requirements tested.
- Security validation completed.
- Results and defects recorded.

6. Test Environment

Item	Environment
Client	Chrome/Edge/Firefox
Server	Development/test backend
Database	Configured test database
API	Postman or equivalent
Version Control	Git/GitHub
Data	Dummy users/admin/complaints

7. Test Cases

ID	Req	Scenario	Procedure	Expected	Type
TC-01	FR-01	Valid registration	Create account with valid data.	Account created.	Functional
TC-02	FR-02	Valid login	Login with correct credentials.	Authenticated and dashboard shown.	Functional
TC-03	FR-02	Invalid login	Use incorrect password.	Login rejected generically.	Functional/ Security
TC-04	FR-03	Submit complaint	Enter valid title/category/description.	Complaint persisted with identifier/status.	Functional
TC-05	FR-04	Missing fields	Submit incomplete complaint.	Submission rejected with validation.	Functional
TC-06	FR-05/06	View/track complaint	Open complaint list and details.	Own records and current status displayed.	Functional
TC-07	FR-07/SEC-02	Unauthorized admin access	Normal user calls admin endpoint.	Request rejected (403/equivalent).	Security
TC-08	FR-08	Admin management	Admin opens and filters	Records displayed and	Functional

ID	Req	Scenario	Procedure	Expected	Type
			complaint list.	filter works.	
TC-09	FR-09	Status update	Admin changes status.	Update persisted and visible.	Functional
TC-10	FR-10	Resolution	Admin enters resolution remarks.	Resolution stored/displayed.	Functional
TC-11	NFR-01	Response time	Measure representative API requests.	95% within 2 seconds.	Non-functional
TC-12	NFR-03	Credential/secret protection	Inspect storage, responses and config.	Passwords hashed; secrets not exposed.	Security

8. Test Schedule

Activity	Planned Timing
Test design	06-Oct-2026
Environment setup	06-Oct-2026
Functional/security execution	06–07-Oct-2026
Defect correction/regression	During sprint
Final summary	After execution

9. Roles and Responsibilities

Role	Responsibility
Team Lead/Product Owner	Scope and acceptance criteria
Test Owner(s)	Design/execute tests and record results
Developers	Fix defects and support test setup
All Members	Review requirements and contribute to testing

10. Risks and Mitigation

Risk	Mitigation
Incomplete implementation	Prioritize high-priority requirements and test incrementally
Environment failure	Keep reproducible setup/test data
Auth defects	Test auth early
Changing requirements	Update SRS, RTM and tests together

11. Suspension and Resumption

- Suspend if environment is unavailable or build blocks most cases.
- Resume after environment restoration and blocking fixes.

12. Test Case Management and Traceability

SRS	Tests
FR-01	TC-01
FR-02	TC-02, TC-03
FR-03	TC-04
FR-04	TC-05
FR-05/06	TC-06
FR-07	TC-07
FR-08	TC-08
FR-09	TC-09

SRS	Tests
FR-10	TC-10
NFR-01	TC-11
NFR-03/04	TC-07, TC-12

13. Test Metrics and Reporting

- Executed/passed/failed percentages
- Defect counts and severity
- Requirement coverage
- Response-time measurements
- Final test summary and unresolved risks

14. Approvals

Role	Name	Signature/Date
Team Lead	<Name>	
Test Owner	<Name>	
Faculty/Reviewer	<Name>	